115. If H is the harmonic mean of numbers $1, 2, 2^2, 2^3, \ldots, 2^{n-1}$, then what is n/H equal to ?

(a) $2-\frac{1}{2^{n+1}}$

(b) $2-\frac{1}{2^{n-1}}$

(c) $2+\frac{1}{2^{n-1}}$

(d) $2-\frac{1}{2^n}$

116. Let P be the median, Q be the mean and R be the mode of observations $x_1, x_2, x_3, \ldots x_n$. Let $S=\sum_{i=1}^{n}(2x_i-a)^2$. S takes minimum value, when a is equal to

(a) P

(b) $\frac{Q}{2}$

(c) $2Q$

(d) R

117. One bag contains 3 white and 2 black balls, another bag contains 2 white and 3 black balls. Two balls are drawn from the first bag and put it into the second bag and then a ball is drawn from the second bag. What is the probability that it is white ?

(a) $\frac{6}{7}$

(b) $\frac{33}{70}$

(c) $\frac{3}{10}$

(d) $\frac{1}{70}$

118. Three dice are thrown. What is the probability that each face shows only multiples of 3 ?

(a) $\frac{1}{9}$

(b) $\frac{1}{18}$

(c) $\frac{1}{27}$

(d) $\frac{1}{3}$

119. What is the probability that the month of December has 5 Sundays ?

(a) 1

(b) $\frac{1}{4}$

(c) $\frac{3}{7}$

(d) $\frac{2}{7}$

120. A natural number n is chosen from the first 50 natural numbers. What is the probability that $n+\frac{50}{n}<50$?

(a) $\frac{23}{25}$

(b) $\frac{47}{50}$

(c) $\frac{24}{25}$

(d) $\frac{49}{50}$

कच्चे काम के लिए जगह